

School Time Table Organizer

User Manual

Project

School Time Table Organizer System

1. Introduction

The School Time Table Organizer System is a web-based application designed to help schools manage:

- Teacher information
- Teacher workload
- Class subject allocation
- Weekly timetable creation
- Class teacher management
- Timetable printing

The system uses:

- HTML
- CSS
- JavaScript
- Firebase Firestore Database

This application helps reduce manual timetable work and avoids duplicate or incorrect timetable entries.

2. Main Modules

The system contains the following modules:

Menu	Purpose
Add Teachers	Add and manage teacher records
Teacher Working Periods per Week	Assign teacher workload

Class Subjects	Define subject periods for each class
Time Table Organizer	Create and manage timetable
Class Teacher List	View all class teachers
Non-Class Teachers List	View non-class teachers
Print Time Table	Print class or teacher timetable

3. System Features

Teacher Management

- Add teacher records
- Edit teacher details
- Delete teacher records
- Search teachers
- Prevent duplicate class teacher assignment

Teacher Workload Management

- Assign:
 - Classes per week
 - Free periods
- Prevent duplicate workload entries

Class Subject Management

- Assign weekly subject periods for every class
- Supports:
 - English
 - Hindi
 - Telugu
 - Maths
 - Science
 - Social
 - Computer

- Biology
- Chemistry
- Physics
- Lab Periods
- PT

Timetable Creation

- Create timetable using:
 - Teacher
 - Class
 - Subject
 - Day
 - Period
- Supports:
 - Class Period
 - Free Period

Validation Features

The system automatically checks:

- Subject availability
- Teacher workload balance
- Duplicate class teacher assignment
- Empty subject allocation
- Remaining subject periods

Printing Features

- Print:
 - Class Timetable
 - Teacher Timetable
 - Class Teacher List
 - Non-Class Teacher List

4. Workflow of the System

Step 1 – Add Teachers

Open:

Add Teachers

Fill:

- Teacher Name
- Subject
- Class & Section
- Is Class Teacher?

Buttons:

Button	Function
Add	Save new teacher
Edit	Update selected teacher
Delete	Remove selected teacher

Important Rules

- Teacher name automatically converts to uppercase.
- One teacher can be class teacher for only one class.
- Duplicate class teacher assignment is blocked.

5. Teacher Working Periods per Week

Open:

Teacher Working Periods per Week

Fill:

- Select Teacher
- Number of Classes per Week
- Number of Free Periods

Purpose

This module controls:

- Teacher workload
- Available periods
- Free periods

Validation

- Duplicate workload entries are not allowed.

6. Class Subjects Module

Open:

Class Subjects

Purpose

Define how many periods each subject requires per week for a class.

Example

Subject	Weekly Periods
English	6
Maths	7
Science	5

Features

- Add subject allocation
- Edit subject allocation
- Delete subject allocation

Validation

- Duplicate class subject records are not allowed.

7. Time Table Organizer

Open:

Time Table Organizer

This is the main module of the project.

Step-by-Step Timetable Creation

Step 1 – Select Teacher

Choose a teacher from the dropdown.

System automatically displays:

- Subjects handled
- Classes handled
- Class teacher details
- Remaining classes/week
- Remaining free periods

Step 2 – Select Class

Choose class & section.

System automatically loads:

- Class Subject Summary

Step 3 – Select Subject

Choose subject for allocation.

Step 4 – Select Work Type

Options:

- Class Period
- Free Period

Step 5 – Select Days

Choose:

- MON
- TUE
- WED
- THU
- FRI
- SAT

Step 6 – Select Periods

Choose:

- Period 1 to Period 8

Step 7 – Create Timetable

Click:

Create Time Table

The system will:

- Save timetable
- Reduce teacher workload count
- Reduce subject remaining count
- Refresh timetable grids

8. Timetable Validation Rules

The system prevents invalid timetable entries.

Validation 1 – Subject Required

If subject is empty:

Please select subject

Validation 2 – Teacher Required

If teacher is not selected:

Select teacher

Validation 3 – Workload Limit

If selected periods exceed teacher availability:

Not enough available periods for this teacher

Validation 4 – Missing Class Subject Summary

If class subject data is not available:

Class Subject Summary not found!

Validation 5 – Subject Count Reached Zero

If subject periods are exhausted:

Selected subject has no remaining periods!

9. Timetable Display

The system automatically displays:

Class Timetable

Shows:

- Subject
- Teacher
- Day
- Period

Teacher Timetable

Shows:

- Class
- Subject
- Assigned periods

10. Delete Selected Periods

Teachers can:

- Select timetable cells

- Delete selected periods

When deleted:

- Teacher workload is restored
- Subject count is restored

11. Class Teacher List

Open:

Class Teacher List

Displays:

- Serial Number
- Teacher Name
- Assigned Class

Features

- Load list
- Reload list
- Print list

12. Non-Class Teacher List

Open:

Non-Class Teachers List

Displays:

- Teacher Name
- Subject
- Classes handled

Features

- Load list
- Reload list
- Print list

13. Print Time Table

Open:

Print Time Table

Supports:

Print Class Timetable

- Select class
- Click:
 - Show Class TT
 - Print

Print Teacher Timetable

- Select teacher
- Click:
 - Show Teacher TT
 - Print

14. Database Collections Used

The system uses Firebase Firestore collections.

Collection	Purpose
tt_teachers	Teacher records
tt_teacher_periods	Teacher workload
tt_class_subjects	Subject allocation
tt_timetable	Timetable entries

15. Technologies Used

Technology	Purpose
HTML	Page structure
CSS	Styling
JavaScript	Logic
Firebase Firestore	Cloud database

16. Important System Logic

Automatic Features

- Auto uppercase teacher names
- Auto workload deduction
- Auto subject count reduction
- Auto timetable refresh
- Auto duplicate checking

Safety Features

- Prevent negative subject count
- Prevent duplicate class teachers
- Prevent empty timetable entries
- Prevent invalid workload allocation

17. Best Practices for Users

Recommended Order

First:

Add Teachers

Second:

Add Teacher Working Periods

Third:

Add Class Subjects

Fourth:

Create Timetable

Fifth:

Print Timetable

18. Troubleshooting

Problem	Solution
Teacher not appearing	Add teacher first
Timetable not saving	Check subject allocation
Subject count zero	Increase subject periods
Print empty	Load timetable before printing
Duplicate class teacher	Assign another teacher

19. Advantages of the System

- Fast timetable generation
- Reduced manual errors
- Automatic validations
- Easy printing
- Easy teacher management
- Cloud database support
- User-friendly interface

20. Conclusion

The School Time Table Organizer System is a complete timetable management solution for schools. It simplifies timetable preparation, teacher workload management, subject allocation, and timetable printing through an easy-to-use web interface integrated with Firebase Firestore.

The system improves:

- Accuracy
- Speed
- Timetable organization
- Administrative efficiency